

S SRI RAM

Electronics & Communication Engineering Student | Embedded Systems & IoT
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CAREER OBJECTIVE

Final-year Electronics and Communication Engineering student specializing in embedded systems, IoT, and hardware design, with four end-to-end projects spanning autonomous robotics, real-time power monitoring, and voice-controlled systems. Proficient in Embedded C, Verilog HDL, PCB design, and Arduino/ESP32 development. Seeking a Project Engineer Trainee - Hardware role to apply hands-on firmware and hardware debugging skills to production-grade embedded products.

SKILLS

Embedded & Hardware: Embedded C, Arduino, ESP32, ESP32-CAM, Verilog HDL, PCB Design, KiCad, Cadence Virtuoso

Programming & Tools: Python (Basic), MATLAB (Basic), C (Basic)

PROJECTS

Smart Waste Cleaning Robot - ESP32-CAM | Academic Project

April 2026

- Engineered an autonomous ESP32-CAM waste-collection robot combining real-time object detection with obstacle avoidance for unmanned water-surface navigation.
- Built the motor-control and navigation firmware in Embedded C, enabling the robot to detect, approach, and collect floating waste without manual steering.

Smart 3-Phase Power Monitoring & Automatic Phase Selector - ESP32 | Academic Project

March 2026

- Developed an ESP32-based IoT system for real-time 3-phase voltage and current monitoring with automatic relay-based phase switching to maintain uninterrupted power supply.
- Designed a live cloud dashboard using ThingSpeak and Vercel, and validated the complete control logic through Wokwi simulation before deploying to physical hardware.

Voice-Controlled Robot - VC-02 Module | Academic Project

February 2026

- Built a voice-controlled robot using Arduino and the VC-02 recognition module, enabling wireless navigation through spoken commands.
- Implemented L298N motor-driver control logic in Arduino IDE (C/C++) to translate recognized voice commands into real-time motor actions.

ESP32-Based Temperature & Humidity Monitoring System - ESP32 | Academic Project

June 2025

- Developed an ESP32-based environmental monitoring system using a DHT11 sensor and 0.96-inch OLED display for real-time temperature and humidity measurement and visualization.
- Designed the KiCad schematic and power supply section using an LM2596 buck converter, integrating sensor interfacing, I2C communication, and regulated power management.

EXPERIENCE

Industrial Intern - Telecommunication Department

June 16 - June 29, 2026

NLC India Limited, Neyveli

- Gained hands-on exposure to industrial telecommunication systems and optical fiber network infrastructure.
- Worked with EPABX systems and LAN networking, strengthening troubleshooting and technical documentation skills.

VLSI Intern

Dec 15 - Dec 19, 2025

MAXVY Technologies

- Simulated digital logic circuits and counters using Verilog HDL to reinforce core digital design concepts.
- Studied the FPGA and VLSI design flow end-to-end, from RTL design through simulation and synthesis basics.

Embedded Systems Intern

May 2025 - June 2025

Bevy Wise Networks, Tirunelveli

- Programmed microcontrollers and interfaced sensors as part of hands-on embedded systems training.
- Contributed to basic IoT and automation projects, gaining practical exposure to hardware connections and debugging workflows.

EDUCATION

B.E. - Electronics and Communication Engineering

CGPA: 7.72 / 10

Einstein College of Engineering, Tirunelveli

Higher Secondary Education

72%

Vivekananda Vidhyashram Matric Higher Secondary School, Tirunelveli

CERTIFICATIONS

- MATLAB Onramp - MathWorks
- Verilog Language & Applications - Cadence Training
- Introduction to C - SoloLearn
- Arduino Design and Code - Udemy
- PCB Design using KiCad